Goxpyriment: A Go Framework for Behavioral and Cognitive Experiments

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Abstract

We introduce `goxpyriment`, a new open-source software framework for programming behavioral and cognitive experiments using the Go programming language. The library is designed to address some limitations of existing Python-based experiment tools, particularly the runtime environment complexity that frequently complicates deployment across laboratories. Because Go is a compiled language that can natively embed assets (e.g., graphics, audio files, and stimulus lists), `goxpyriment` compiles entire experiments into single, self-contained executable binaries with zero runtime dependencies. This drastically simplifies distribution to collaborators and testing computers. `goxpyriment` prioritizes timing reliability. Go's garbage collector can be disabled, highly-reducing the probability of unpredicable pauses, latencies that can ruin the timing of the experiment. Keyboard and mouse events' timestamps at the operating system level are available, allowing one to avoid continuous pooling to get precise reaction times. The programming interface, inspired by Expyriment (Krause & Lindemann, 2014), was designed be human friendly. The library includes an array of visual stimuli (text, shapes, images, Gabor patches, motion clouds, ...) and audio capabilities (WAV playback and tone generation). A set of 35 classic psychology experiments implemented in goexpriment are provided that promote not only learning by humans but also boost the ability of modern AI-assisted coding tools to program experiments. Finally, a migration guide from other frameworks is provided. The framework is released under the GNU General Public License v3 and is freely available at https://github.com/chrplr/goxpyriment.

Keywords: framework, library, stimulus presentation, timing, behavioral experiments, psychophysics, psychology

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Introduction

Experiment-control software occupies a central position in behavioral and cognitive neuroscience: the temporal precision of stimulus delivery and response recording, the ergonomics of experiment programming, and the ease of deploying experiments to laboratory hardware each affect the quality and reproducibility of empirical data. Over the past two decades, Python has become the dominant language for experiment programming, primarily through two libraries: Expyriment (Krause & Lindemann, 2014) and PsychoPy (Peirce et al., 2019). Before Python, MATLAB with the Psychophysics Toolbox (Brainard, 1997; Kleiner et al., 2007) was the de facto standard in many laboratories. More recently, web-based frameworks such as jsPsych (de Leeuw, 2015; de Leeuw et al., 2023) and lab.js (Henninger et al., 2022) have enabled large-scale online data collection.

Each class of tool involves tradeoffs. MATLAB, with hooks to custom written C/C++ libraries (e.g. the Psychtoolbox), offers good timing on well-configured hardware but is proprietary and expensive. Psychtoolbox scripts written on Windows often do not work out of the box on Linux (and vice-versa). Python tools provide an open, flexible programming environment with a large scientific ecosystem, but they introduce two practical difficulties. First, deploying an experiment to a computer that does not already have a compatible Python environment requires package management (conda, pip, venv) and can be a source of breakage across operating-system versions. For example, installing the psychopy python module under Linux often crashes while attempting to compile the wxPython package when there is no precompiled wheel. Second, Python's garbage collector can introduce unpredictable pause times that corrupt the timing of stimulus presentation and response recording during the critical trial loop. Web-based tools trade timing precision for accessibility: browser-mediated stimulus presentation is subject to event-loop latency that make sub-frame timing difficult or impossible (Anwyl-Irvine et al., 2021; Bridges et al., 2020).

The Go programming language (Donovan & Kernighan, 2016) offers a different point in the design space. It was designed for simplicity and has much fewer moving parts than Python; as a consequence, the code tends to be more "boring" (there is only one way to do things) but predictable. Because Go is strongly typed, and favors error checking, developping code typically requires less run-time debugging. Go compiles to native machine code for each target platform and produces self-contained binaries with no runtime dependencies. Its garbage collector is concurrent and can be suspended for the duration of a timing-critical section. These properties make it an attractive substrate for a behavioral experiment library that prioritizes timing reliability and deployment simplicity.

The present article describes the design and capabilities of `goxpyriment`, discusses its timing architecture, and situates it relative to existing tools. Specifically, we highlight three main contributions of this framework: (1) zero-dependency deployment via single compiled binaries that simplify distribution across laboratory hardware; (2) native high-precision timing achieved directly through the Go and SDL3 APIs without requiring external C/C++ helper engines; and (3) a linear, straightforward API that is highly suited for modern AI-assisted coding workflows.

Core Advantages of a Compiled Framework

The choice of Go as the implementation language for `goxpyriment` provides several critical advantages over the interpreted languages traditionally used in behavioral research.

Single-Binary Deployment (Zero Dependencies)

Go compiles program into a single statically linked executable with no external runtime dependencies. Crucially, the `go-sdl3` package embeds the pre-built SDL3 runtime libraries (SDL Developers, 2026) directly into the executable. The stimuli, experimental lists, fonts, ... can be embedded too. As a results, distributing an experiment requires copying just one file; there is no need to manage virtual environments, package versions, or interpreter installations across laboratory computers. This substantially reduces the operational overhead of running experiments on multiple machines or sharing materials with collaborators.

Explicit Garbage Collection Control

While the garbage collectors in languages like Python or JavaScript can introduce unpredictable pause times that corrupt timing, Go exposes a runtime API for suspending the garbage collector. `goxpyriment` uses this mechanism in its timing-critical loops, ensuring that stimulus onsets and offsets are not delayed by memory management tasks.

An AI-Friendly API

Go's design emphasizes simplicity and predictability. Because the language has few synonymous constructs and a strict type system, it is particularly well-suited for use with AI-powered coding assistants. The linear, straightforward API of `goxpyriment` allows researchers to leverage these assistants to bridge gaps in their programming experience, with the Go compiler providing a safety net by catching many common errors before the experiment is even run.

Timing Architecture and Precision

Hardware-Synchronized Presentation

While many Python-based experiment frameworks achieve high-precision timing by relying on external C/C++ engines like the Psychophysics Toolbox (PTB), `goxpyriment` achieves equivalent low-level control directly through the native SDL3 API (SDL Developers, 2026). `screen.Update()` calls SDL's `SDL_RenderPresent`, which blocks until the display's vertical retrace when VSYNC is enabled. Reliance on SDL3 also provides the possibility to control *variable refresh rate* monitors (although the goxpyriment API itself does not currently wrap this SDL3 functionality.)

High-Precision Streams

For paradigms with dense, high-speed stimulus sequences (e.g., Retinotopic stimulation), `goxpyriment` provides specialized stream functions like `PresentStreamOfImages`. These functions disable garbage collection, drain the event queue, and busy-wait at the sub-frame level to ensure that each stimulus onset is scheduled accurately. A `TimingLog`

records the target and actual onsets for post-hoc verification, as well as the timing of keys pressed during the presentation of the stream.

Hardware-Level Response Timestamps

Response timing in `goxpyriment` leverages SDL3 hardware event timestamps. The `WaitKeysEventRT` method returns the `KeyboardEvent.Timestamp` field, which is populated by the OS input subsystem at hardware-interrupt time (nanoseconds since SDL initialization). By comparing this with the `Screen.FlipNS()` timestamp—which uses the same clock—reaction times can be measured with minimal jitter and no interpreter overhead. This unified timing regime is abstracted through the `ResponseDevice` interface, which explicitly signals whether a timestamp is "precise" (hardware-level) or "non-precise" (polling-based).

The Timing Verification Suite

The repository includes a dedicated timing verification suite (`tests/Timing-Tests/`) that allows researchers to characterize their hardware's performance. The suite includes tests for measuring frame-interval jitter, verifying single-frame flash reliability via photodiode, and calculating audio-visual SOA latencies. Other tests measure hardware-buffer pipeline latency for audio and characterize the USB-serial jitter of trigger devices. Other tests could be implemented upon request by users to the first author. A special folder in goxpyriment's github repository, namely 'performance-reports', is meant for users to share the results of tests on their own hardware.

Lifecycle and Safety

Every experiment centers on an `Experiment` object that manages the window, renderer, and audio subsystems. Execution is typically wrapped in a `Run` function that provides a safety layer: if the participant presses the Escape key or closes the window, the library catches the resulting internal signal, safely flushes the data file to disk, and shuts down the SDL subsystems. This ensures that data is preserved even if a session is aborted prematurely.

Defining Trials and Data Logging

Experiments follow a hierarchical structure of blocks and trials directly inspired expyriment. Each trial is a collection of factors that can be randomized and logged. Data collection is handled by an `exp.Data` object, which writes to a CSV file with extensive metadata headers.

Pre-Experiment Setup Dialog

`goxpyriment` provides an optional graphical `GetParticipantInfo` dialog that opens before the main experiment window (See Fig. 1) It allows the experimenter to input subject IDs, select experimental conditions from discrete menus, and toggle settings like fullscreen mode. These values are automatically recorded in the output data file header.

Programs that use `GetParticipantInfo` also accept a `-headless` command-line flag. When present, the dialog is skipped entirely and field defaults (or values previously

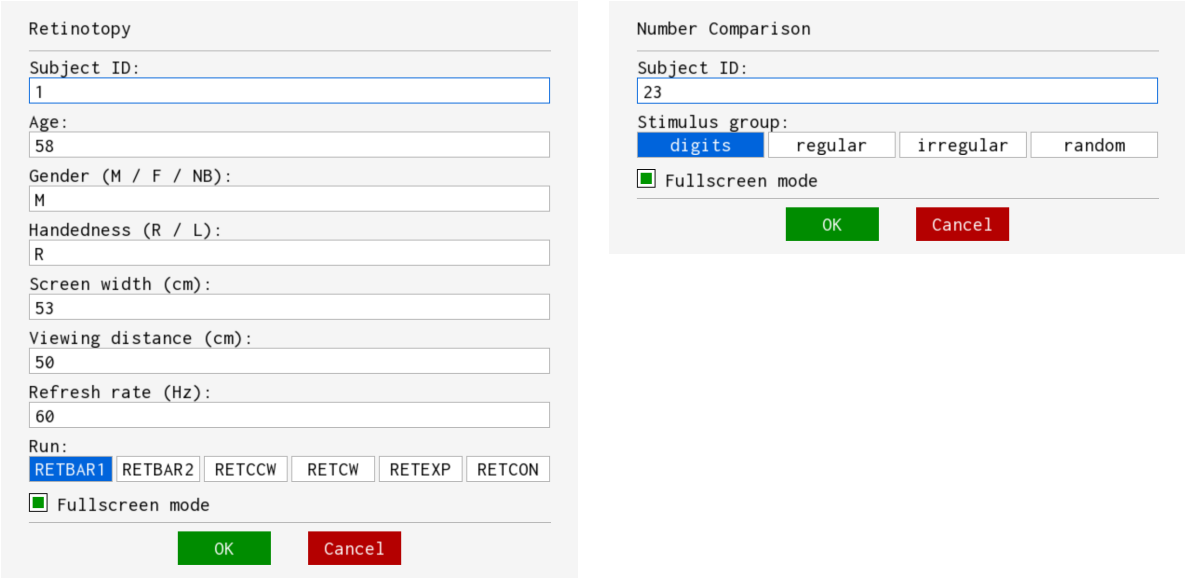

Figure 1

Examples of `GetParticipantInfo` *dialogs for the* Retinotopy *(left) and* Number-Comparison *(right) paradigms provided in the examples accompanying the library. These UIs makes it practical for the experimenter or participant to choose among a small fixed set of options without typing, which reduces setup errors when launching an experiment from a desktop shortcut or file manager. Note that which fields appear in the UI is configurable.*

saved to the cache) are used without opening any window. This is convenient for scripted or automated runs (e.g. continuous integration checks or batch stimulus generation) where interactive input is not possible.

Comparison with Existing Tools

Table 1 summarizes key properties of `goxpyriment` alongside four widely used experiment frameworks.

Table 1. Comparison of selected experiment frameworks.

Property	goxpyriment	Expyriment	PsychoPy	jsPsych
Language	Go	Python	Python	JavaScript
Execution	Native binary	Interpreter	Interpreter	Browser
Deployment	Single binary	pip/conda env	Standalone app or pip	Node.js or CDN
GC control	Explicit disable	No	No	No
VSYNC sync	Yes (SDL3)	Yes (Pygame/SDL2)	Yes (Pyglet/PTB)	Limited (rAF)
Hardware triggers	Yes	Yes	Yes	No

Property	goxpyriment	Expyriment	PsychoPy	jsPsych
Response device API	Yes	Device-specific	Device-specific	N/A
Event timestamp source	SDL3 hardware	Python `time.time()`	iohub or Python clock	`performance.now()`
Stimulus set	Moderate	Moderate	Extensive	Moderate
GUI builder	No	No	Yes (Builder)	No
Online deployment	No	No	Partial (PsychoJS)	Yes
Reference	This paper	(Krause & Lindemann, 2014)	(Peirce et al., 2019)	(de Leeuw, 2015)

`goxpyriment` resembles Expyriment in its API design: a central experiment object, a `present()`-based stimulus API, a structured trial-loop model, and output to `.csv`-format files with metadata. The primary technical differences are the use of Go rather than Python and the use of SDL3 rather than the older Pygame/SDL2 stack. From the user point of view, goexpyriment adds 'streams' objects which, as their name indicate, allow to display streams of stimuli, withal tightly controlled timing.

PsychoPy offers a substantially larger stimulus set and a graphical Builder environment; `goxpyriment` has neither of these. However, PsychoPy's greater complexity also makes deployment and version management more challenging. jsPsych enables large-scale online data collection, which `goxpyriment` does not support. The MATLAB Psychophysics Toolbox (PTB) offers the most mature timing validation in the field; `goxpyriment` aims to achieve comparable precision through its native compiled architecture.

Example Experiments

The repository provides over forty examples demonstrating classic paradigms, including the Stroop task, Attentional Blink (RSVP), Masked Priming, Retinotopic mapping for fMRI, and adaptive threshold estimation using the QUEST algorithm. These examples serve as both functional templates and a gallery of the framework's capabilities, ranging from simple parity-judgment tasks to complex moving-dot attention studies.

Limitations and Future Directions

`goxpyriment` is currently at the beta-stage, that is, the API is stabilized and we are waiting for bug reports to address them.

The current library covers the stimuli most commonly needed in cognitive psychology but does not yet include complex 3D stimuli (which can nevertheless be programmed directly using the underlying SDL api), nor has built-in eye-tracker support. We plan to add this in a future version.

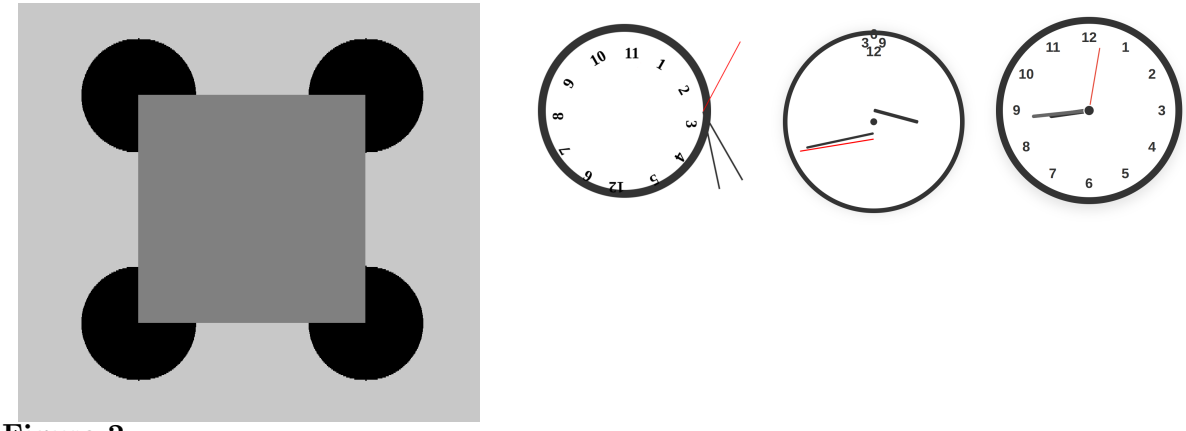

Figure 2

AI "blunders": *Left: Asked by the first author to write a program displaying Kanizsa's illusory rectangle, an AI produced the result above, explaining: "I have changed the Kanizsa example to make things robust and more visible [...] A darker gray rectangle is drawn at the center, clearly visible against the light-gray background.". Right: Output from the AI clock project where AIs are asked to write programs drawing a clock; note the correct response on the rigth (See https://clocks.brianmoore.com/ for more).*

A cautionary note about AI-assisted coding for psychologists

As mentioned above, goexpyriment is well-suited for AI assisted coding as its API and the provided examples provide strong guidance for AI coding agents prompted to program psychology experiments. The most recent experiments in the examples gallery were actually entirely generated by Claude or Gemini (see the section Authors' contributions below. Yet, even if these agents are extremely apt and quick at coding, human supervision is still clearly necessary as exemplified by Fig. 2.

Author Contributions

OB**Christophe Pallier**: Conceptualization, Software implementation, Documentation and Paper Writing, with assistance from **Claude Sonnet 4.6** and **Gemini 2.5**. Consistent with current editorial norms (Nature Portfolio, 2023; Springer Nature, 2023), Claude and Gemini are listed under Author Contributions rather than as a named author, because authorship requires the capacity to take accountability for the work. Christophe Pallier takes full responsibility for all content.

Julie Bonnaire performed timing tests and redacted a report available in https://github.com/chrplr/goxpyriment/tree/mainperformance-reports

Marie-France Fourcade designed and built the photodiode system used to measure visual presentation delays in some tests.

Open Practices Statement

The source code of `goxpyriment` is available under the GNU General Public License v3 at https://github.com/chrplr/goxpyriment. All example experiments described in this article are included in the repository. No empirical data are reported in this article.

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appendix

Code Availability and Deployment

`goxpyriment` is available at https://github.com/chrplr/goxpyriment under the GPLv3 license. While the single-binary format simplifies distribution, users should be aware of modern operating system security features.

Windows: SmartScreen warning. When a user downloads a `.exe` file from the internet and tries to run it, Windows Defender SmartScreen may display a "Windows protected your PC" warning, because the executable has not been digitally signed. To proceed the user clicks **More info** and then **Run anyway**. This prompt appears only on first launch and only for files with the "downloaded from the internet" mark of origin; files copied from a USB drive or an internal network share bypass the check.

macOS: Gatekeeper. macOS includes a security system called **Gatekeeper** that checks whether applications have been signed with an Apple-issued certificate and submitted to Apple for notarization (a process that requires a paid Apple Developer account, $99/year). Applications distributed outside the Mac App Store that are not notarized receive a quarantine attribute when downloaded, which triggers Gatekeeper on first launch.

Two dialogs may appear, depending on how the user attempts to open the application.

1. **"App is damaged" dialog** (Figure 3). This alarming message appears when double-clicking an unsigned `.app` bundle. Despite the wording, the application has not been corrupted. **Workaround:** right-click (or Control-click) the `.app`, choose **Open** from the contextual menu, then click **Open** in the confirmation dialog. The exception is remembered; subsequent double-clicks launch the app normally.
2. **Privacy & Security panel** (Figure 4). After a blocked launch attempt, macOS records the blocked app in **System Settings → Privacy & Security**. Scrolling to the blocked-app entry and clicking **Open Anyway** (followed by the administrator password) has the same effect as the right-click method.

For plain command-line binaries (not `.app` bundles), the right-click workaround does not apply. Instead, the quarantine attribute can be removed with:

```
xattr -d com.apple.quarantine ./myexperiment
chmod +x ./myexperiment
./myexperiment
```

Linux. Linux binaries compiled with `go build` require no installation: copy the binary to the target machine, mark it executable (`chmod +x`), and run it. For broader distribution, the AppImage format (https://appimage.org) encapsulates a binary and its required shared libraries into a single portable file that runs on most Linux distributions without root access.

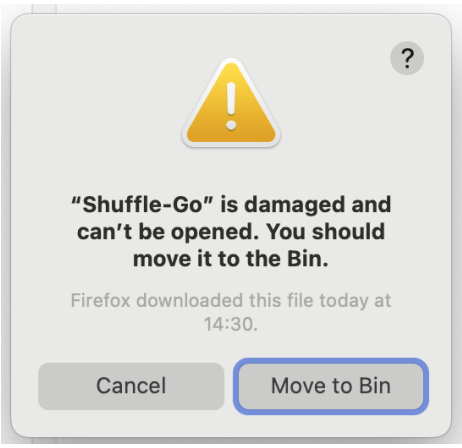

Figure 3

The macOS Gatekeeper "damaged" dialog that appears when double-clicking an unsigned `.app` *bundle downloaded from the internet. The application is not corrupted; the message reflects the absence of an Apple notarization certificate. Right-clicking the app and selecting* Open, *then confirming in the second dialog, bypasses the restriction permanently for that binary. Note that sometimes it is also necessary to run a 'xttr -d com.apple.quarantine myapp' command (see text)*

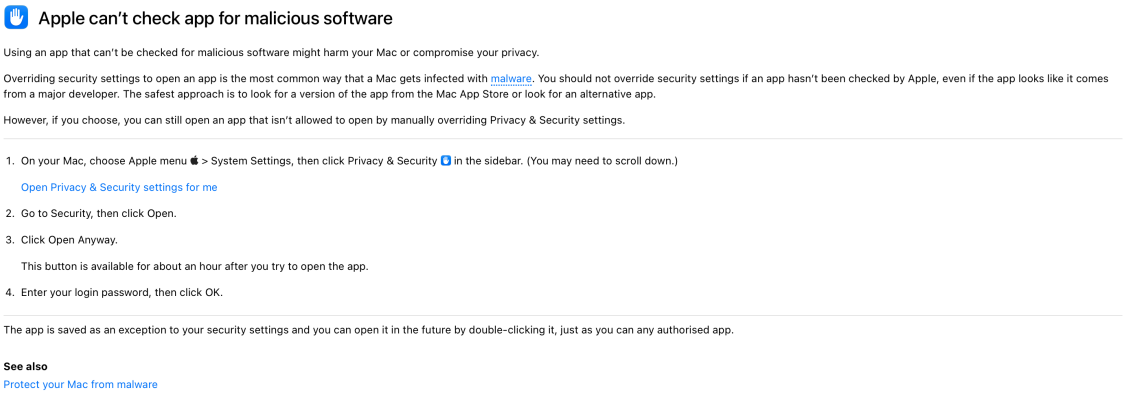

Figure 4

The Privacy & Security pane in macOS System Settings showing a blocked unsigned application. Clicking Open Anyway *and supplying the administrator password grants a permanent exception for that binary.*